

Post-Hospital Discharge Risk Recommendation System

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Background

- Hospitals are penalized under CMS HRRP for excess readmissions
- Identifying high-risk hospitals is critical for improving outcomes
- Existing approaches focus on prediction without actionable insights

Data

- CMS HRRP hospital-level dataset
- Aggregated to hospital-level features

Features

- avg_err - readmission performance
- err_std - variability
- total_discharges - patient volume
- measure_count - reporting coverage
- stability_z - performance consistency

Methodology

Models evaluated:

- Logistic Regression
- Random Forest
- Gradient Boosting
- XGBoost

Evaluation:

- ROC-AUC
- Cross-validation

Target Definition

High Risk = volume-weighted ERR > 1.0 (ERR > 1 indicates excess readmissions compared to expected CMS benchmark)

Hospital readmission risk can be reliably assessed using hospital-level performance features, and feature-driven recommendations enable targeted improvement strategies.



Post-Hospital Discharge Risk Recommendation System

This dashboard assesses hospital readmission-related risk using engineered hospital-level performance features and provides targeted recommendations to support improvement planning.

Choose Assessment Mode

How would you like to assess a hospital?

☒ Select Existing Hospital

☐ Manual Input

Select a Hospital

Choose a hospital from the processed dataset

ADENA REGIONAL MEDICAL CENTER

Selected Hospital Overview

Hospital: ADENA REGIONAL MEDICAL CENTER	State: OH	Volume-Weighted ERR: 0.985
Facility ID: 360159	Observed Risk Label: Low Risk	Avg ERR: 0.989

Hospital Profile Inputs

Average ERR (avg_err)	ERR Variability (err_std)	Measure Count
0.9886	0.0765	6

Average ERR Z-score (avg_err_z)

Total Discharges

Stability Z-score (stability_z)

-0.1367

1127

0.5341

Run Risk Assessment

Risk Assessment Summary

Risk Probability	Model Risk Classification	Risk Tier
0.249	Low Risk	Low Risk

This hospital shows relatively low readmission-related risk.

What This Risk Level Means

Low Risk means the hospital appears relatively stable based on the model and historical feature patterns. This does not mean zero risk, but it suggests that current performance is closer to expected or manageable levels.

Why the Hospital May Be Receiving This Risk Level

- The current feature pattern appears relatively stable and does not show strong high-risk signals.

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Recommended Next Steps

Monitoring Focus

- Current performance appears relatively stable. Maintain existing interventions and continue periodic monitoring.

Hospital Profile Used for Assessment

	avg_err	avg_err_z	err_std	total_discharges	measure_count	stability_z
0	0.9886	-0.1367	0.0765	1127	6	0.5341

Reference Hospital Record

	facility_name	state	avg_err	avg_err_z	err_std	total_discharges	measure_count	stability_z	volume_weighted_err	high_risk
1963	ADENA REGIONAL MEDICAL CENTER	OH	0.9886	-0.1367	0.0765	1127	6	0.5341	0.9849	0

About this dashboard

Interactive dashboard demonstrating hospital selection, risk assessment, and recommendation generation workflow

Model Results

Model	ROC-AUC	Accuracy	F1 Score
Random Forest (Tuned)	0.947	0.876	0.886
Random Forest (Base)	0.947	0.878	0.889
Gradient Boosting	0.942	0.874	0.887
XGBoost	0.941	0.864	0.876
Logistic Regression	0.940	0.870	0.883

All models achieved similar performance (ROC-AUC ~0.94–0.95); tuned Random Forest selected based on stability and cross-validation performance

Key Insight

- Minimal performance gap across models
- Strong predictive signal driven by feature engineering
- Problem is structured and partially linearly separable

Risk Interpretation

- High Risk - excess readmission patterns
- Moderate Risk - early warning signals
- Low Risk - relatively stable performance

Recommendation Framework

- Recommendations are feature-driven and based on hospital performance patterns
- Not fixed per risk level
- Adapt to hospital-specific feature values

Examples

- High avg_err - improve discharge planning
- High variability - standardize workflows
- High volume - strengthen care coordination

Conclusion

- Combines prediction, explanation, and recommendation
- Transforms model output into a decision-support system

